TENDERPROOF

Engineering evidence before equipment release

An engineering review workbench for requirements, supplier decisions and shipment evidence.

SwissChivers / Hack Apertus / Track 2B / October 2026

Can we buy? Can we ship?

Proposed users: EPC package engineers, procurement leads and quality managers

Package engineer	Procurement lead	Quality manager
Which clauses apply? Which interpretation was reviewed?	Which supplier facts and documented costs support the decision?	Does this evidence apply to this unit and this revision?

Keep the source, checks and decision authority together.

Four connected workspaces

Implemented review surfaces share one versioned package

Workspace	What the reviewer does	What stays connected
Requirements	Review clauses and typed proposals	Source page, span and interpretation
Supplier comparison	Inspect technical outcomes and documented costs	Supplier configuration and source facts
Acceptance	Review evidence applicability and release prerequisites	Unit / revision / evidence gaps
History and approvals	Inspect changes, role approvals and immutable packets	Reviewed snapshot and authority

The current real package contains genuine inference; all results remain pending.

A proposal becomes a governed decision

End-to-end mechanism implemented in the prototype

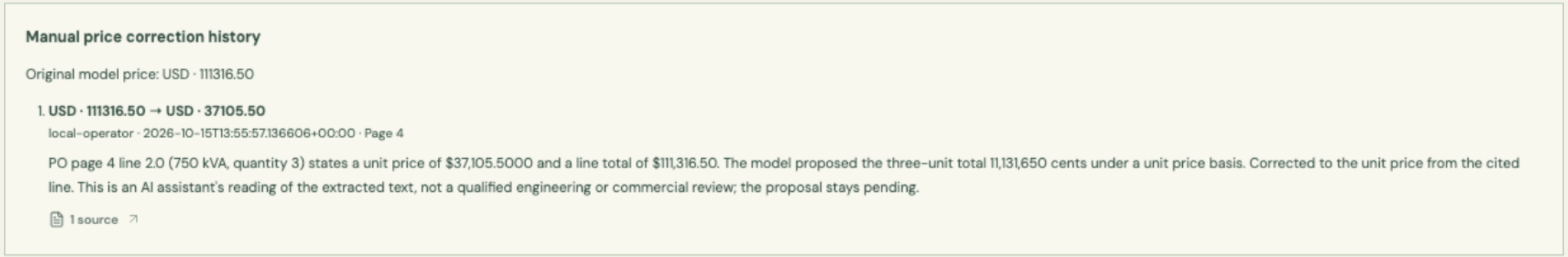
01	02	03	04	05
Intake	**Interpret**	**Review**	**Check**	**Authorize**
Docling spans Original hash	Apertus proposes Pending candidates	Human meaning Source references	Z3 constraints Exact arithmetic	Role approvals Current snapshot

A content change recomputes the package and makes prior approvals stale.

Unresolved extraction conditions block both release gates.

A real transformer purchase

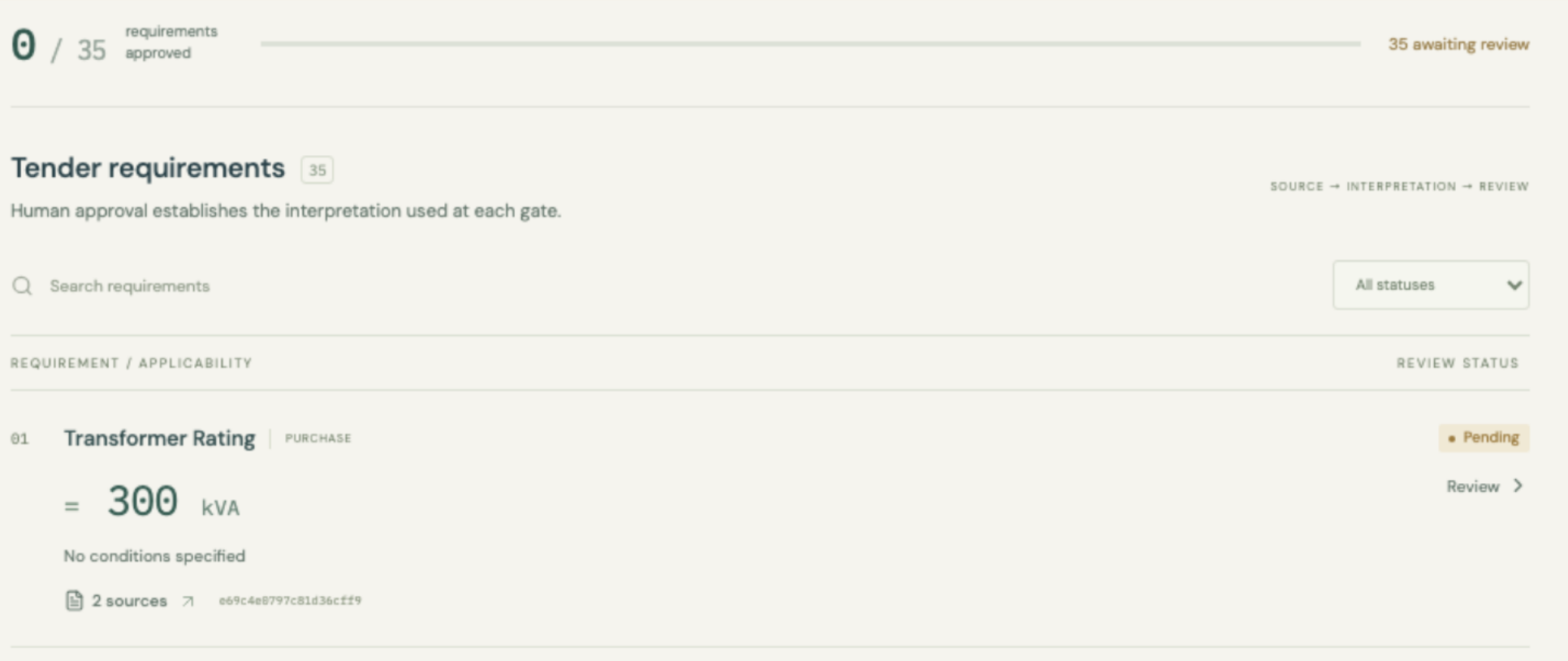
OP34276 item 2 / three 750 kVA units / recorded line USD 111,316.50



PO unit price corrected with a source citation; original model output preserved.

Actual Apertus output needs review

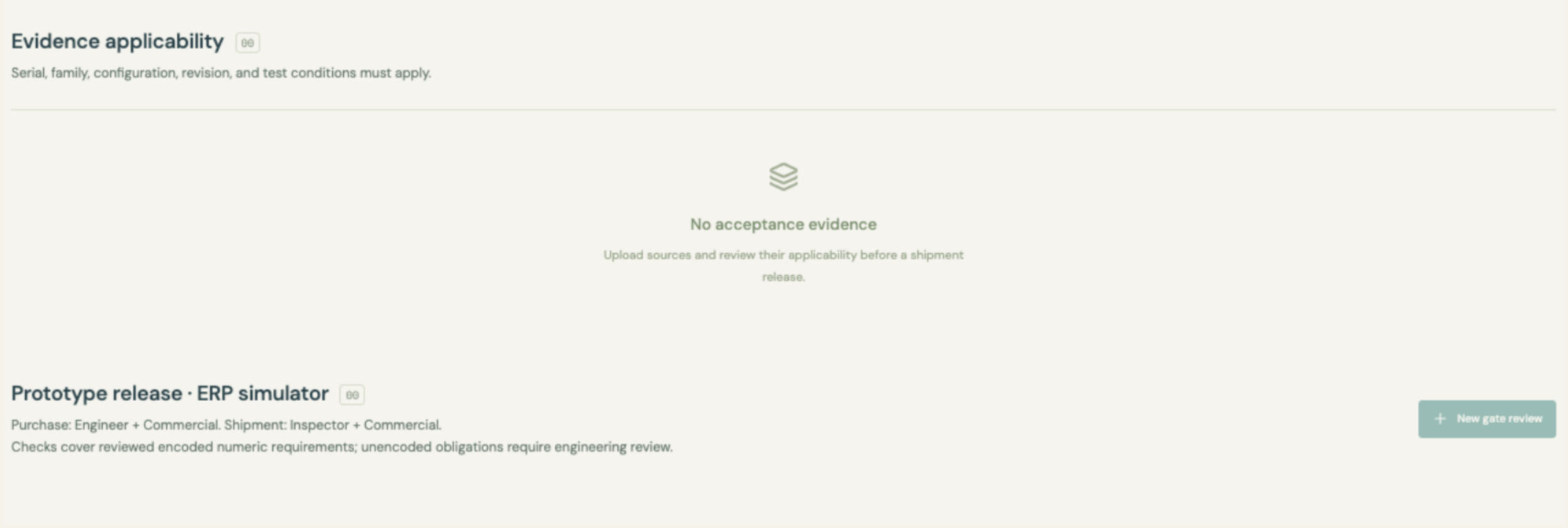
35 pending requirements / 4 pending offers / no approvals



swiss-ai/Apertus-v1.5-8B / 427.12s API wall time / full tender

Missing evidence blocks acceptance

Incomplete extraction and missing FAT remain visible at the gate



Acceptance stays unknown without applicable FAT, serial evidence and approved drawings.

Designed for organizational control

On-premise target / authenticated GPU execution / durable application state

React
Four review workspaces

→

FastAPI
Roles, checks, snapshots

↔

Apertus
Structured proposals

↕

PostgreSQL
Snapshots and audit

Docling parsing
Temporal activities
Scoped MCP access

Z3 + exact arithmetic inside the API

Enter through one real engineering package

Proposed business wedge: enterprise / project deployment for equipment procurement

Start	Validate	Expand
One equipment package	Expert-reviewed decisions	Additional project teams
A named review owner	Corrections and effort	Equipment schemas
Existing source documents	Deployment / operating cost	Enterprise integrations

Adjacent market: Mercanis raised >$20m Series A (June 2025).

Next commercial test: a design partner with a held-out real package.

Measured behavior and remaining work

Evidence for the implementation, and the next validation step

Observed

330 backend / 39 frontend tests
5 completed real inference runs
64 preserved call traces
Snapshot-bound release controls

Still to validate

Held-out model / semantic quality
Engineering expert decisions
Customer workflow and economics
Production recovery / security

A real source starts the review. Release still needs applicable proof.

Appendix / Twenty planned extensions

Sequenced expansion hypotheses; none is a current-demo integration

1 / Integrate	2 / Broaden engineering	3 / Follow execution	4 / Learn and optimize
05 ERP purchase orders	10 Switchgear / protection	15 Manufacturing hold points	20 Lifecycle energy losses
06 PLM synchronization	11 Battery / inverter schemas	16 Shipping documentation	21 Supplier outcome analytics
07 Supplier clarifications	12 BIM / IFC asset links	17 Site acceptance	22 Spare interchangeability
08 Corporate SSO / authority	13 Simulation adapters	18 Contract change orders	23 Authorized expert learning
09 Cross-project packages	14 Licensed standards library	19 Warranty investigations	24 Cross-project sourcing

Sequence depends on validated demand, source rights and engineering review.

Appendix / Recorded source details

Actual item-2 price tabulations and the governing clarification

Bidder as written	Basic USD/unit	Alternate USD/unit
R&D Marketing	37,105.50	Blank
Benson Guam Enterprises Inc.	25,600.00	22,845.00
Polyphase System Inc.	27,252.26	Item-2 blank
JMI Edison	34,250.00	Blank

Amendment II: flip-top response “no.” Delivery: 24 weeks after approved drawings.

Source readings require review; price alone does not establish compliance or TCO.